

S.No	Architecture / Component	Technology Chosen	Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript, Bootstrap	Creates a responsive and user-friendly interface for farmers to enter data and view recommendations.
2	Backend / Server-Side	Python, Flask	Processes user requests, performs prediction, and connects the ML model with the web application.
3	Database / Data Storage	CSV Dataset, Pandas	Stores and manages crop, soil, and weather data used for training and prediction.
4	Machine Learning	Scikit-learn (Random Forest), NumPy	Trains and predicts the best crop based on soil and weather conditions.
5	Cloud / Hosting / Deployment	Flask (Localhost), Render	Runs and deploys the application for online access.
6	Version Control	Git, GitHub	Tracks code changes and supports collaboration among team members.
7	Third-Party APIs / Tools	OpenWeather API	Fetches real-time weather data to improve crop recommendation accuracy.

Summary

The Opti Crop system integrates **HTML, CSS, JavaScript, Bootstrap, Python, Flask, Scikit-learn, Pandas, NumPy, GitHub, Render, and OpenWeather API** to provide an intelligent, reliable, and scalable crop recommendation solution for farmers.